CONTINUITY

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Name: Solutions

1. Show that the given function is continuous at the given number, a.

(a) $f(\cancel{x}\ t) = \dfrac{t^2+5t}{2t+1}$, $a = 2$.

$$f(2) = \frac{2^2+5(2)}{2(2)+1} = \frac{4+10}{4+1} = \frac{14}{5}$$

$$\lim_{t \to 2} \frac{t^2+5t}{2t+1} = \frac{2^2+5(2)}{2(2)+1} = \frac{4+10}{4+1} = \frac{14}{5} = f(2)$$

So f is continuous.

(b) $f(x) = 3x^4 - 5x + \sqrt[3]{x^2+4}$, $a = 2$.

$$\lim_{x \to 2} f(x) = 3(2)^4 - 5(2) + \sqrt[3]{2^2+4} = 3(16) - 10 + \sqrt[3]{8}$$
$$= 48 - 10 + 2 = 40$$

$$f(2) = 3(2)^4 - 5(2) + \sqrt[3]{2^2+4} = 40 = \lim_{x \to 2} f(x).$$

So f is continuous.

2. Show that the function

$$f(x) = \frac{x-1}{3x+6}$$

is continuous on the interval $(-\infty, -2) \cup (-2, \infty)$.

There is a discontinuity at $x=-2$ because $3(-2)+6=-6+6=0$.

When $a \neq 2$,

$$\lim_{x\to a} \frac{x-1}{3x+6} = \frac{\lim_{x\to a}(x-1)}{\lim_{x\to a}(3x+6)} = \frac{a-1}{3a+6} = f(a)$$

So f is continuous on $(-\infty,-2) \cup (-2,\infty)$.

3. Find the number k that makes the function

$$f(x) = \begin{cases} \dfrac{x^3-8}{x^2-4} & x \neq 2 \\ k & x = 2 \end{cases}$$

continuous.

$$\lim_{x\to 2} \frac{x^3-8}{x^2-4} = \lim_{x\to 2} \frac{(x-2)(x^2+2x+4)}{(x-2)(x+2)} = \lim_{x\to 2} \frac{x^2+2x+4}{x+2} = \frac{4+4+4}{4} = \frac{12}{4} = 3.$$

To make f continuous we need

$$k = f(2) = \lim_{x\to 2} \frac{x^3-8}{x^2-4} = \boxed{3}$$

4. Use continuity to evaluate the given limit.

(a) $\lim_{x\to\pi} \sin\left(x + \sin(x)\right)$

$$= \sin\left(\lim_{x\to\pi}(x+\sin(x))\right) = \sin\left(\lim_{x\to\pi} x + \lim_{x\to\pi}\sin(x)\right)$$

$$= \sin(\pi + \sin(\pi)) = \sin(\pi+0) = \sin(\pi) = \boxed{0.}$$

(b) $\lim_{x\to 4} x\sqrt{20-x^2}$

$$= \left(\lim_{x\to 4} x\right)\left(\lim_{x\to 4}\sqrt{20-x^2}\right) = \lim_{x\to 4} x\sqrt{\lim_{x\to 4}(20-x^2)} = 4\sqrt{20-4^2} = 4\sqrt{4} = 4(2)$$

$$= \boxed{8}$$

5. Use the Intermediate Value Theorem to show that there is a solution to the given equation in the specified interval.
Note: You do not need to find the solution!

(a) $x^4 + x - 3 = 0$, $(1, 2)$

$1^4+1-3 = 1+1-3 = 2-3 = -1 < 0$

$2^4+2-3 = 16+2-3 = 15 > 0$

So by the IVT, there is some value c such that $1<c<2$ and

$c^4+c-3=0.$

(b) $\frac{2}{x} = x - \sqrt{x}$, $(2, 3)$

It is equivalent to find a solution to $x-\sqrt{x}-\frac{2}{x}=0$ on $(2,3)$

$2-\sqrt{2}-\frac{2}{2} = 2-\sqrt{2}-1 = 1-\sqrt{2} < 0$

$3-\sqrt{3}-\frac{2}{3} = \frac{9}{3}-\frac{2}{3}-\sqrt{3} = \frac{7}{3}-\sqrt{3} > 0$

So there exists some c such that $2<c<3$ and

$\frac{2}{c} = c-\sqrt{c}.$

(c) $\cos(x) = x$, $(0, 1)$

$\cos(0)-0 = 1-0 = 1 > 0$

$0<1<\pi/2$, so $0=\cos(\pi/2) < \cos(1) < 1 = \cos(0)$

$\Rightarrow -1 < \cos(1)-1 < 0$

So by the IVT, there exists some c such that

$0<c<1$ and $\cos(c)=c.$